

Multi-task Learning enhanced CNN-LSTM Framework for Weather Spatiotemporal Prediction and Occlusion based Explainable Analysis

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Atmospheric variables have multi-dimensional and complex nonlinear relationships, which impose challenges for the comprehensive and reliable implementation of multi-dimensional weather data in spatial and temporal dimension predictions for weather forecasting. Therefore, this work adopted a multi-task learning enhanced convolutional neural network (CNN) and long short-term memory network (LSTM) to simultaneously predict atmospheric pressure and temperature fields. Specifically, the constructed model applied CNN and LSTM to extract spatial and temporal features from historical weather data, encoded and learned common representations applicable to both tasks, and shared and transferred relative knowledge based on the multi-task learning structure, coupled with task-specific components that decoded and specialized in predicting each variable. On this basis, an explainable method based on occlusion approach is used to analyze the importance of individual elements in the weather forecast result generation process. Through control experiments, it can be preliminarily proved that the multi-task learning strategy can be beneficial to predict accuracy and explainability. This work is expected to provide a reference and insight into more complex and multi-dimensional data, comprehensive integration, and reliable deep learning prediction in the future.

Multi-task learning | CNN-LSTM | Occlusion based explainable | Spatiotemporal weather forecasting

Introduction

Weather forecasting is a critical scientific endeavor with profound impacts on fields such as agriculture, transportation, and daily human activities. Traditional numerical weather prediction methods mainly focus on employing physics-based models with complex differential equations to describe atmospheric dynamics. However, these methods are highly dependent on domain knowledge of meteorology and constantly face challenges due to uncertainty quantification and complex nonlinear relationships (1). In recent years, with the development of computing resources and artificial intelligence, data-driven methods, especially deep learning, have gained increasing attention in the field of weather forecasting due to their ability to handle inherent correlations in datasets without explicitly encoding physical laws and principles (2).

Among the deep learning models, Convolutional Neural Networks (CNNs) have proven their efficiency for spatial pattern recognition in meteorological fields (3), while recurrent architectures like long short-term memory networks (LSTMs) are also widely applied for modeling temporal dependencies in fields such as weather sequences (4). Based on this, researchers further explored the combination of CNN and LSTM for atmospheric dynamics prediction. For example, Chen et al. introduced three-dimensional convolutional neural networks (3DCNN) and two-dimensional convolutional neural networks (2DCNN) together to better understand the spatial relationships of typhoon formation features and applied LSTM to examine the time series of relationships during typhoon development (5). Hou et al. also integrated CNN and LSTM for hourly temperature prediction, proving their efficiency especially for long time series of hourly temperature data (6).

Despite this, conventional deep learning methods mainly focused on predicting a single weather variable separately (7). However, atmospheric variables are inherently interrelated, and changes in one variable often affect other variables through complex feedback mechanisms; at the same time, changes in each variable are essentially caused by changes in the same atmospheric system. The prediction tasks of multiple variables that belong to the same system are essentially close tasks (8), for which knowledge sharing and implicit learning of the relationship between variables can improve the effect of the prediction task in many aspects, including accuracy and reliability.

Multi-task learning (MTL) provides a strategy for deep learning to learn multiple close tasks simultaneously and share knowledge between tasks to improve generalization (9). Therefore, MTL can be used to predict multiple atmospheric variables simultaneously, allowing the model to capture relationships between variables that may be missed when predicting each variable individually, which makes it suitable for weather variable predictions. Therefore, the MTL strategy has also been adopted in atmospheric and geologic variables analysis. For example, Xie et al. proposed a novel end-to-end CNN-based Multi-task Weather Recognition Network with multi-scale weather cues, to obtain the description of weather conditions by jointly tackling

the weather-cue segmentation task and weather classification task (10). Et al. proposed a spatiotemporal deep-learning-based hydrological model that couples the 2-Dimension CNN and LSTM to analyze the meteorological and hydrological data (11). However, the application of MTL with CNN-LSTM to weather forecasting remains relatively limited explored, particularly for simultaneous prediction of atmospheric pressure and temperature fields.

Furthermore, although deep learning methods have demonstrated impressive predictive capabilities in data-driven climate forecasting, their decision-making and result-generating processes are opaque. This essential black box makes it difficult to ensure their reliability, imposing a major challenge on scientific research and applications. Therefore, network explainable methods have been proposed and improved in recent years, which can be divided into two main categories, which are intrinsic interpretability and post-hoc explainable methods (12). In terms of meteorological applications, McGovern et al. reviewed various interpretable AI methods, emphasizing their potential for enhancing scientific understanding and operational forecasting (13). Toms et al. applied gradient-based attribution methods to climate models to identify regions of importance for El Niño predictions (14). However, due to the complexity of gradient problems, time dependencies that LSTM may cause, and the huge computing resources required for some explainable methods, conventional and straightforward methods such as Layer-wise Relevance Propagation (LRP) and SHapley Additive exPlanations (SHAP) may lead to an unreliable result, or consume huge computing resources.

This work introduces a multi-task deep learning framework for simultaneously predicting sea level pressure and 2-meter temperature fields within a specific area. For spatial feature extraction and temporal sequence modeling, CNNs and LSTMs are combined as a CNN-LSTM network, which enables the capture of both spatial patterns and temporal dynamics. On this basis, the model architecture incorporates shared layers that encode and learn common representations applicable to both tasks, as well as task-specific components that decode and specialize in predicting each variable. Based on this, this work further conducted a occlusion based explainable analysis about the result generation process of the network. According to the author's investigation, there is still relative limited research about the combination of MTL and CNN-LSTM for weather data dimension analysis. Besides, due to the complex of networks and computation consumption, explainable strategies for LSTM-based weather spatial and temporal dimension result analysis is also limited. Therefore, this work is expected to provide an insight into related and more complex and multi-dimensional data comprehensive integration and reliable deep learning prediction in the future.

Methodology

In order to apply the MTL structure to achieve simultaneous prediction of future temperature and pressure data based on past continuous historical data, and to perform efficient reliability analysis, this work adopt two main techniques: MTL-enhanced CNN-LSTM network, as well as occlusion-based explainable strategy specifically for computation-resource-constrained application.

MTL-enhanced CNN-LSTM deep learning network Architecture. The proposed multi-task model architecture consists of two main components: shared encoding CNN-LSTM part and task-specialized decoding parts, as shown in Fig. 1.

Spatial Feature Extraction with CNN encoder. The CNN encoder is responsible for learning spatial features from the input data, which includes pressure and temperature variables as separate inputs. the model contains a series of convolutional modules to extract hierarchical spatial patterns from the consecutive previous data. Among them, multiple CNN layers can have the potential to gradually reduce the spatial dimensions of the feature map and simultaneously learn more abstract representations of the input data. ReLU activation function is applied after each convolution operation to introduce nonlinearity to the model, enabling it to learn more complex patterns. Batch normalization is applied in each module to ensure that the network can learn more effectively by stabilizing the activation values between different mini-batches. Finally, the applied max-pooling layer can be beneficial reduce the spatial resolution of the input data by selecting the maximum value from the adjacent value pool, reducing the dimensionality and computational resource requirements of the network. The principle of the CNN module can be expressed as:

$$y_{i,j} = \sum_{m=-k}^k \sum_{n=-k}^k x_{i+m,j+n} \cdot w_{m,n} + b \quad [1]$$

$$\text{ReLU}(x) = \max(0, x) \quad [2]$$

$$y_{i,j} = \max_{m,n} x_{i+m,j+n} \quad [3]$$

where $y_{i,j}$ is the output pixel at position (i, j) , x is the input image, w is the filter weight, b is the bias term, and k represents the size of the kernel.

The output of the CNN encoder is a set of feature maps that represent the spatial characteristics of both pressure and temperature. These feature maps are then flattened into a one-dimensional vector, which is passed into the LSTM network for further processing.

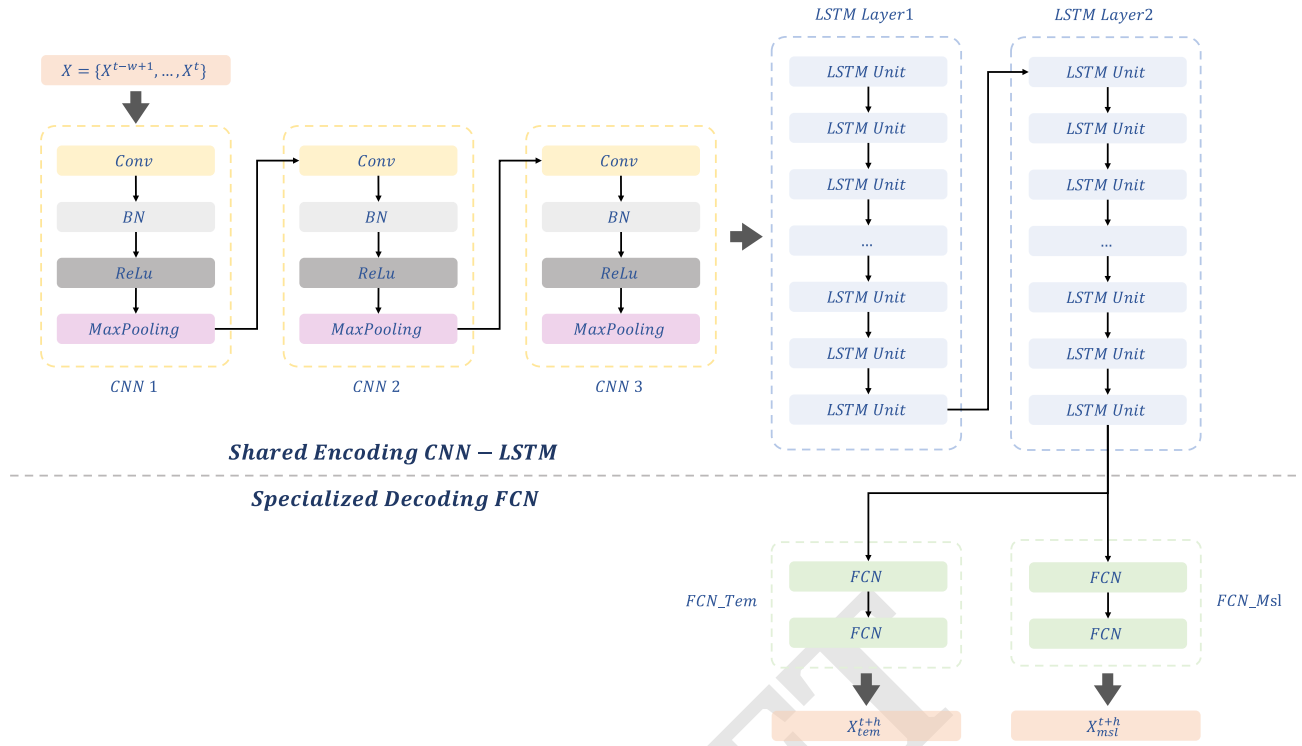


Fig. 1. The structure of constructed MTL enhanced CNN-LSTM

Temporal Sequence Learning with shared LSTM encoder. The LSTM network is applied to capture the temporal dependencies within the data. Given a sequence of input vectors x_t , the LSTM processes each element of the sequence using the following standard LSTM equations:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad [4]$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad [5]$$

$$g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g) \quad [6]$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad [7]$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot g_t \quad [8]$$

$$h_t = o_t \cdot \tanh(c_t) \quad [9]$$

where: i_t, f_t, g_t, o_t are the input, forget, cell, and output gates at time step t , c_t is the cell state, and h_t is the hidden state at time step t .

The LSTM network is designed to retain long-term dependencies in the data through its memory cell, allowing it to process sequences effectively. The final hidden state h_T (from the last time step T) is then passed to the task-specific output layers.

Multi-Task Output Layers with specialized decoding FCN. Based on the features extracted through the shared encoding part, the multi-task learning structure allows the model to simultaneously decode both temperature and pressure information using separate output layers for each task. The output layers for each task are fully connected network (FCN). For pressure prediction, the output $\hat{y}_o$ is obtained through the following fully connected layers:

$$\hat{y}_o = \text{FCN}_o(\text{ReLU}(\text{FCN}_{o1}(h_T))), \quad o \in \{\text{pressure, temperature}\} \quad [10]$$

where FCN_{o1} and FCN_o are the fully connected layers for pressure, and $\hat{y}_o$ represents the predicted results.

On this basis, the multi-task learning framework will ensure that the shared CNN and LSTM components learn for two tasks simultaneously, thereby improving model performance by leveraging common representations for temperature and pressure.

Occlusion-based explainable strategy for Weather Prediction Explainability. Recurrent architectures often suffer from vanishing or exploding gradients, especially on longer sequences, which often makes gradient-based attribution methods such as LRP unreliable. Conventional SHAP methods can be computationally inefficient due to computational resource constraints. Therefore, this work adopts an occlusion-based importance analysis method that provides a model-independent view of feature correlations. This strategy systematically perturbs input features to directly measure their impact on predictions, providing a robust alternative that does not rely on gradient computation and is therefore applicable to any black-box model.

The fundamental principle behind the applied occlusion-based approach is to systematically mask portions of the input and observe the resulting change in prediction (15). For each spatial location (i, j) in the input domain, the importance score can be computed as:

$$I(i, j) = |f(x) - f(x_{i,j}^{\text{occ}})| \quad [11]$$

where $f(x)$ is the original prediction, and $f(x_{i,j}^{\text{occ}})$ is the prediction when a window centered at (i, j) is occluded. In this work, occlusion is performed using a sliding window of size $w \times w$ with stride s , systematically moved across the input field:

$$x_{i,j}^{\text{occ}}[i : i + w, j : j + w] = 0 \quad [12]$$

The magnitude of prediction change directly represents the importance of the occluded region. Larger deviations indicate features with greater influence on the model's prediction. This approach provides an intuitive measure of feature importance that is conceptually accessible and requires no knowledge of the model's internal architecture.

For the specific multi-variable weather prediction task, the importance of pressure and temperature fields at each timestep in the input sequence is calculated, respectively. The implementation uses a window-based approach with parameters adjustable for the desired granularity and can be adaptive based on computational efficiency:

$$\text{for } t \in \{1, \dots, T\}, c \in \{\text{pressure, temperature}\}, (i, j) \in \text{Grid} : \quad [13]$$

$$I_t^c(i, j) = |f(x) - f(x_{t,c,i,j}^{\text{occ}})| \quad [14]$$

where T is the number of timesteps, and $x_{t,c,i,j}^{\text{occ}}$ represents the input with occlusion applied to channel c at timestep t and spatial location (i, j) . Based on this, the resulting importance maps are further normalized to facilitate visualization and comparison:

$$I_{\text{normalized}} = \frac{I - \min(I)}{\max(I) - \min(I)} \quad [15]$$

This strategy is expected to provide several advantages especially when the network model is complex and computational resource is limited. Besides, through the paralleling the process across spatial locations and strategically selecting window sizes and strides, the applied occlusion strategy can achieve an efficient importance computation, which providing insights into the spatiotemporal dependencies captured by the model, by quantifying the specific importance of geographical regions and variables in specific location forecasts at specific timesteps.

Case Study

Dataset and Preprocessing. Given the provided historical dataset of sea level pressure and 2-meter temperature over a sequence of time steps, the weather forecasting is essentially a spatiotemporal prediction task to predict both variables at the next time step. Formally, let X_p^t and X_T^t represent the pressure and temperature fields at time step t , respectively. Given the provided historical dataset, $\{X_p^{t-w+1}, \dots, X_p^t\}$ and $\{X_T^{t-w+1}, \dots, X_T^t\}$ which consists of gridded fields with dimensions including time, latitude, and longitude, covering a specific geographical region, a window size w can be determined to predict X_p^{t+h} and X_T^{t+h} , where h is the forecast horizon.

In order to guarantee the effectiveness of gradient descent for different tasks and to facilitate the measurement of the effect of model training, the normalization is conducted by subtracting the mean and dividing by the standard deviation:

$$X_{\text{normalized}} = \frac{X - \mu}{\sigma} \quad [16]$$

where μ and σ are the mean and standard deviation of the variable over the entire dataset.

Based on this, a time series dataset has been constructed to form input-output pairs for sequence prediction:

$$X = \{X^{t-w+1}, \dots, X^t\}, \quad y = \{X_{\text{tem}}^{t+h}, X_{\text{msl}}^{t+h}\} \quad [17]$$

where w is the window size and h is the forecast horizon. In this work, the $w=7$, while $h=1$, which means this work will predict the temperature and pressure data for the next day based on seven consecutive days of meteorological data. Based on this, the time series dataset is randomly split into training (80%) and testing (20%) sets.

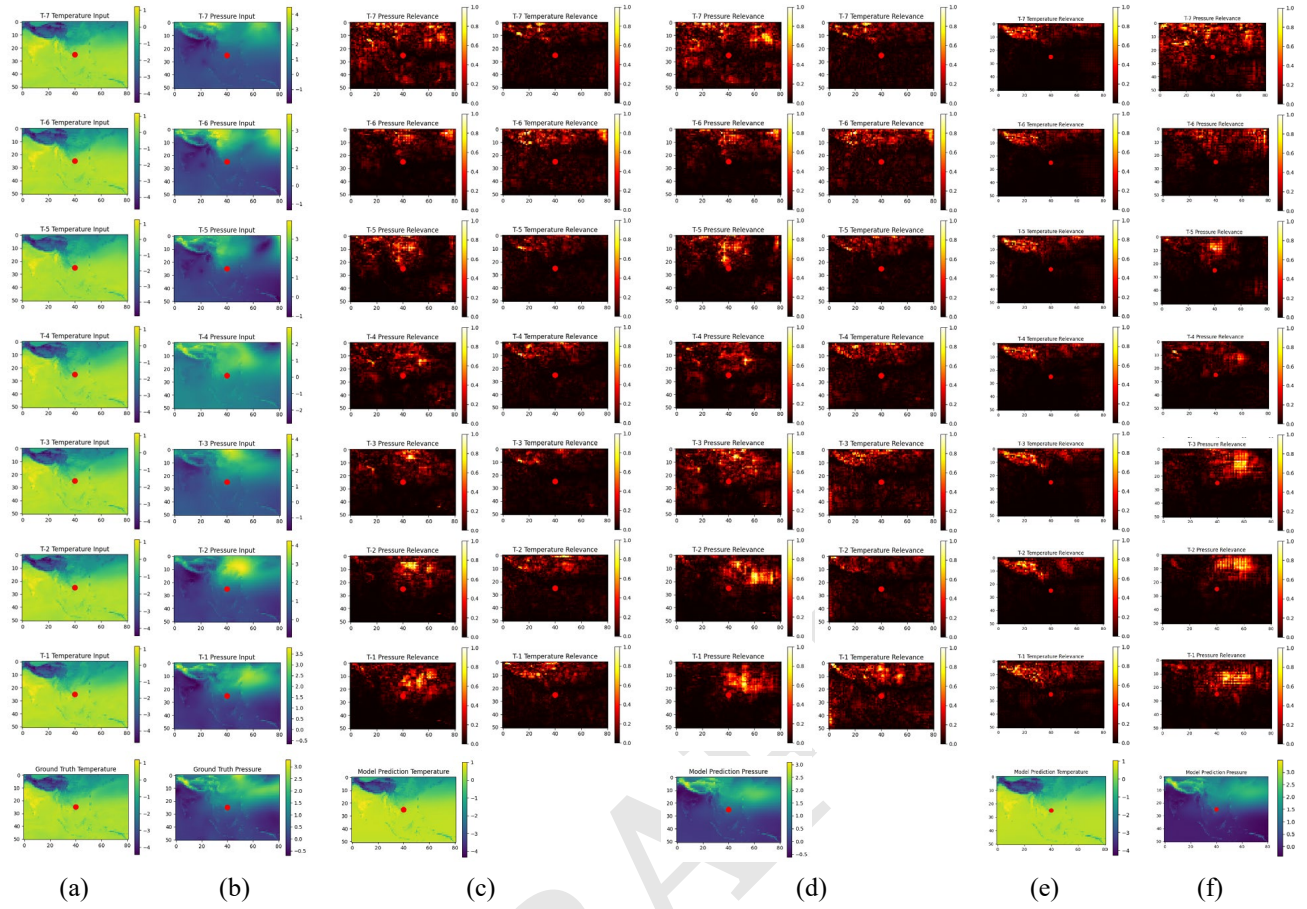


Fig. 2. Occlusion based explainable analysis result comparison of MTL-enhanced and separate predictions: (a) time series temperature data, (b) time series pressure data, (c) temperature prediction process and result by MTL structure, (d) pressure prediction process and result by MTL structure, (e) temperature prediction process and result by conventional structure, (f) pressure prediction process and result by conventional structure

training settings. This work applied a multi-task loss function, which combines the losses from both temperature and pressure predictions. The loss function is a weighted sum of the Mean Squared Error (MSE) losses for each task:

$$\mathcal{L}_{\text{total}} = \lambda_p \mathcal{L}_p + \lambda_T \mathcal{L}_T \quad [18]$$

where $\mathcal{L}_p$ and $\mathcal{L}_T$ are the mean squared error (MSE) losses for pressure and temperature predictions, respectively, and λ_p and λ_T are weights that balance the importance of each task, which is 0.5 in this case.

In addition, in subsequent comparisons, this work will also introduce the metric of generalization error E_{gen} to measure the generalization ability between different models:

$$\text{Generalization Error}(E_{\text{gen}}) = \frac{\|\text{Validation Loss} - \text{Training Loss}\|}{\text{Training Loss}} \times 100\% \quad [19]$$

For training process, the Plateau strategy is applied to dynamically adjust the learning rate of machine learning to help the model get rid of the local minimum and fine-tune when learning encounters difficulties. All programmings are executed on NVIDIA GeForce RTX 4060 Laptop GPU-8GB. All training sets the same random number seed to ensure the reproducibility of the conclusion.

Verification of the effectiveness of MTL structure. In order to verify the effectiveness of the multi-task learning architecture and its improvement in explainable performance, this work sets up two control groups based on the same CNN-LSTM architecture, with the spatial and temporal dimensions unchanged: predicting temperature based on temperature data alone, and predicting air pressure based on pressure data alone. Among them, for the task of temperature prediction, delete the $\text{FCN}_{\text{pressure}}$, while for the task of air pressure prediction, delete the $\text{FCN}_{\text{temperature}}$.

Result Analysis. Based on previous settings, the results are shown as Table 1 and Fig. 2. It can be seen from Table 1 that, from the perspective of R-squared and MSE alone, both the traditional CNN-LSTM and the MTL enhanced CNN-LSTM

Table 1. Statistical Parameters and Measurements

	MTL-temp	MTL-pre	Temperature	Pressure
MSE	0.0176	0.0859	0.0134	0.0813
R^2	0.9822	0.9122	0.9864	0.9178
E_{gen}	1.73%	11.1%	1.47%	25%
Time	3079		2791	2580

have achieved high levels of accuracy and fitting capabilities. At the same time, when other conditions remain unchanged, the application of MTL does not have a significant impact on accuracy.

However, since the MTL architecture adopts a shared CNN-LSTM encoding structure, the network can share and migrate underlying features, which greatly improves the network training efficiency and reduce the overall time consumption. At the same time, it can be seen that the E_{gen} of the CNN-LSTM based on the MTL architecture has been greatly improved for the prediction of pressure. This shows that under the circumstance of without using dropout and additional regularization, the training of the network based on the MTL architecture and its own generalization performance have been improved to a certain extent.

At the same time, it can be seen from Fig. 2 that the prediction using the MTL architecture may be able to better consider the time series characteristics during the pressure data processing. For example, for the same pressure prediction, the conventional CNN-LSTM is more concerned with the size of the value and focuses on the pressure data around the prediction point, as shown in 2(f); while the prediction process of the CNN-LSTM based on the MTL architecture shows that due to the introduction of temperature data, it tends to pay more attention to more global air pressure data, as shown in Fig. 2(d). This may be one of the reasons for the poor generalization performance of the ordinary CNN-LSTM.

Unlike the pressure data, the prediction of temperature data shows another change pattern. Compared with the conventional CNN-LSTM, the regional distribution of the CNN-LSTM based on the MTL architecture for historical temperature data has not changed significantly, while its proportion has changed accordingly, as shown in Fig. 2(c) and Fig. 2(e). Combined with the fact that its accuracy has not changed significantly, the reliability of the prediction process and results of the CNN-LSTM based on the MTL architecture for temperature changes analysis has been further consolidated and increased.

Conclusion

In this work, a MTL-enhanced CNN-LSTM framework was constructed for weather spatiotemporal prediction, specifically for predicting sea level pressure and 2-meter temperature fields based on given dataset. The proposed framework leveraged the power of CNNs for spatial feature extraction and LSTM networks for temporal sequence learning. By integrating these models in a multi-task learning structure, the network was able to jointly predict pressure and temperature, facilitating the sharing of learned representations between the two tasks. This architecture allowed for a more efficient model that better captured the interdependencies between atmospheric variables, improving prediction accuracy and model generalization.

Additionally, an occlusion-based explainability method was introduced to provide insight into the decision-making process of the model. This method helped identify the importance of specific spatial and temporal features in the prediction process, offering valuable information for understanding the model's behavior and enhancing trust in its predictions.

Experimental results showed that the MTL-enhanced CNN-LSTM framework provided comparable prediction accuracy, as indicated by the R^2 and MSE metrics, and better training efficiency as the decrease of time consuming, compared to separate CNN-LSTM models for each task. Furthermore, the MTL structure demonstrated superior generalization performance, particularly for pressure prediction. The occlusion-based analysis also revealed the ability of the MTL structure to better consider global and temporal dependencies, further solidifying the model's performance.

Overall, the proposed approach offers a promising avenue for weather forecasting and can be extended to other multi-dimensional, spatiotemporal prediction tasks. Future work can explore more advanced explainable AI methods and further refine the MTL architecture to enhance its scalability and predictive capabilities for large-scale real-world weather data.

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Replication of Results Code and data

The data, code and result will be available via link below within a few days after being sorted out.
https://github.com/sooncheer0420/ME5311_Project_Report

References

1. P Bauer, A Thorpe, G Brunet, The quiet revolution of numerical weather prediction. *Nature* **525**, 47–55 (2015).
2. M Reichstein, et al., Deep learning and process understanding for data-driven earth system science. *Nature* **566**, 195–204 (2019).
3. YM Zhang, H Wang, Multi-head attention-based probabilistic cnn-bilstm for day-ahead wind speed forecasting. *Energy* **278**, 127865 (2023).
4. K Venkatachalam, P Trojovský, D Pamucar, N Bacanin, V Simic, Dwfh: An improved data-driven deep weather forecasting hybrid model using transductive long short term memory (t-lstm). *Expert. systems with applications* **213**, 119270 (2023).
5. R Chen, et al., A hybrid cnn-lstm model for typhoon formation forecasting. *Geoinformatica* **23**, 375–396 (2019).
6. J Hou, Y Wang, J Zhou, Q Tian, Prediction of hourly air temperature based on cnn-lstm. *Geomatics, Nat. Hazards Risk* **13**, 1962–1986 (2022).

- 221 7. JA Weyn, DR Durran, R Caruana, Improving data-driven global weather prediction using deep convolutional neural networks on a cubed sphere. *J. Adv. Model. Earth Syst.* **12**, e2020MS002109
222 (2020).
- 223 8. C Sun, et al., Digital-twin-enhanced metal tube bending forming real-time prediction method based on multi-source-input mtl. *Struct. Multidiscip. Optim.* **65**, 296 (2022).
- 224 9. R Caruana, Multitask learning. *Mach. learning* **28**, 41–75 (1997).
- 225 10. K Xie, L Huang, W Zhang, Q Qin, L Lyu, A cnn-based multi-task framework for weather recognition with multi-scale weather cues. *Expert. Syst. with Appl.* **198**, 116689 (2022).
- 226 11. B Li, et al., Improving lstm hydrological modeling with spatiotemporal deep learning and multi-task learning: A case study of three mountainous areas on the tibetan plateau. *J. Hydrol.* **620**, 129401
227 (2023).
- 228 12. C Molnar, *Interpretable machine learning*. (Lulu. com), (2020).
- 229 13. A McGovern, et al., Using artificial intelligence to improve real-time decision-making for high-impact weather. *Bull. Am. Meteorol. Soc.* **98**, 2073–2090 (2017).
- 230 14. BA Toms, EA Barnes, I Ebert-Uphoff, Physically interpretable neural networks for the geosciences: Applications to earth system variability. *J. Adv. Model. Earth Syst.* **12**, e2019MS002002 (2020).
- 231 15. A Ordookhanians, X Li, S Nakandala, A Kumar, Demonstration of krypton: Optimized cnn inference for occlusion-based deep cnn explanations. *Proc. VLDB Endow.* **12**, 1894–1897 (2019).

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